

# Limit Switch

## KXM Series

### Part Number Description

| KXM - ① ②        |                                    |                                            |                              |
|------------------|------------------------------------|--------------------------------------------|------------------------------|
| ① Head Type      | 301 : Push Plunger                 | 726 : Ø50 Adjustable Roller Lever - Rubber |                              |
|                  | 302 : Roller Plunger               | 901 : Spring Rod                           |                              |
|                  | 312 : Cross Roller Plunger         | 902 : Spring Wire                          |                              |
|                  | 702 : Roller Lever                 | 903 : Spring Rod - Plastic Tip             |                              |
|                  | 703 : Adjustable Roller Lever      | 923 : Spring Rod - Metal Tip               |                              |
|                  | 704 : Adjustable Rod Lever         |                                            |                              |
| ② Indicator Lamp | No mark : Standard (No Indication) | D : LED Lamp (12 ~ 24VDC/AC)               | A : Neon Lamp (100 ~ 240VAC) |

### General Specification

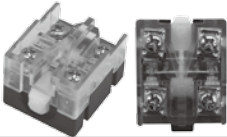
|                       |                                              |                                                 |
|-----------------------|----------------------------------------------|-------------------------------------------------|
| Contact Form          |                                              | 1N/O + 1N/C                                     |
| Contact Material      |                                              | Ag alloy/24K Au Plate                           |
| Rated Current         |                                              | 6A 250VAC · Resistance load                     |
| Insulation Resistance |                                              | Min. 100M $\Omega$ 500VDC Insulation resistance |
| Contact Resistance    |                                              | Max. 25m $\Omega$                               |
| Dielectric Strength   |                                              | 1,500VAC · 50/60Hz 1 Minute                     |
| Internal Voltage      | Between the terminals                        | 1,000VAC · 50/60Hz 1 Min.                       |
|                       | Between each terminal and non-charging parts | 1,500VAC · 50/60Hz 1 Min.                       |
| Life Cycle            | Mechanical                                   | Min. 1,000,000                                  |
|                       | Electrical                                   | Min. 300,000                                    |
| Vibration Resistance  |                                              | 10 ~ 55Hz dual wave length 1.5mm                |
| Shock Resistance      | Malfunction                                  | 98m/s <sup>2</sup> · Max. 10G                   |
|                       | Destruction                                  | 294m/s <sup>2</sup> · Max. 30G                  |
| Degree Of Protection  |                                              | IP66                                            |
| Ambient Temperature   |                                              | -20 ~ +60°C · (with no icing)                   |
| Ambient Humidity      |                                              | Max. 90%RH                                      |
| Weight                |                                              | Approx. 130 ~ 190g                              |

※ Be sure to use a band clip when using the IP66.

| Contact Ratings | Non-Inductive |                |       |           | Inductive  |       |                |       |
|-----------------|---------------|----------------|-------|-----------|------------|-------|----------------|-------|
|                 | Voltage       | Resistive Load |       | Lamp Load | Motor Load |       | Inductive Load |       |
|                 |               | NC(A)          | NO(A) |           | NC(A)      | NO(A) | NC(A)          | NO(A) |
|                 | 125VAC        | 6              |       | 1.5       | 0.7        | 2     | 1              | 3     |
|                 | 250VAC        | 6              |       | 1         | 0.5        | 1.5   | 0.8            | 3     |
|                 | 500VAC        | 2              |       | 0.75      | 0.35       | 0.75  | 0.4            | 0.75  |
|                 | 30VDC         | 5              |       | 3         |            | 3     |                | 4     |
|                 | 125VDC        | 0.4            | 0.2   | -         |            | -     |                | -     |
|                 | 250VDC        |                |       |           |            |       |                |       |

### Contact Block

### Indication Lamp Name Plate

| KXM-101                                                                             | KXM-102                                                                             | KXM-103                                                                               |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 24K Au Plate                                                                        | 12 ~ 24VAC/DC LED Lamp                                                              | 100 ~ 240VAC Neon Lamp                                                                |
|  |  |  |



## Product Selection

|                                                                                     | Indicator Lamp | Part Number | Head Type               | OF                                     | RF                                   | PT    | MD    | OT  | TT    |
|-------------------------------------------------------------------------------------|----------------|-------------|-------------------------|----------------------------------------|--------------------------------------|-------|-------|-----|-------|
|    | Standard       | KXM - 301   | Push Plunger            | 900gf<br>(8.83N)                       | 150gf<br>(1.47N)                     | 1.5mm | 0.7mm | 4mm | 5.5mm |
|                                                                                     | LED Lamp       | KXM - 301D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 301A  |                         |                                        |                                      |       |       |     |       |
|    | Standard       | KXM - 302   | Roller Plunger          | 900gf<br>(8.83N)                       | 150gf<br>(1.47N)                     | 1.5mm | 0.7mm | 4mm | 5.5mm |
|                                                                                     | LED Lamp       | KXM - 302D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 302A  |                         |                                        |                                      |       |       |     |       |
|   | Standard       | KXM - 312   | Cross Roller Plunger    | 900gf<br>(8.83N)                       | 150gf<br>(1.47N)                     | 1.5mm | 0.7mm | 4mm | 5.5mm |
|                                                                                     | LED Lamp       | KXM - 312D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 312A  |                         |                                        |                                      |       |       |     |       |
|  | Standard       | KXM - 702   | Roller Lever            | 600gf<br>(5.88N)                       | 150gf<br>(0.49N)                     | 20°   | 10°   | 75° | 95°   |
|                                                                                     | LED Lamp       | KXM - 702D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 702A  |                         |                                        |                                      |       |       |     |       |
|  | Standard       | KXM - 703   | Adjustable Roller Lever | 800gf<br>(7.84N)<br>~ 342gf<br>(3.35N) | 50gf<br>(0.49N)<br>~ 21gf<br>(0.21N) | 20°   | 10°   | 75° | 95°   |
|                                                                                     | LED Lamp       | KXM - 703D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 703A  |                         |                                        |                                      |       |       |     |       |
|  | Standard       | KXM - 704   | Adjustable Rod Lever    | 800gf<br>(7.84N)<br>~ 230gf<br>(1.99N) | 50gf<br>(0.49N)<br>~ 12gf<br>(0.12N) | 20°   | 10°   | 75° | 95°   |
|                                                                                     | LED Lamp       | KXM - 704D  |                         |                                        |                                      |       |       |     |       |
|                                                                                     | Neon Lamp      | KXM - 704A  |                         |                                        |                                      |       |       |     |       |

# Limit Switch

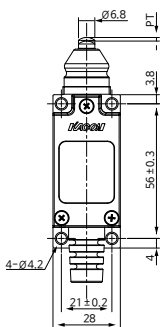
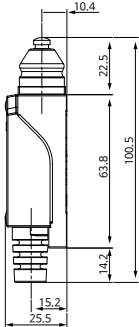
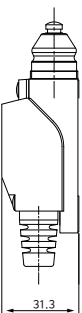
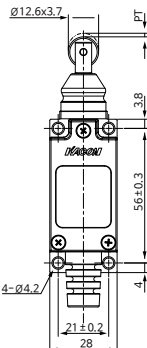
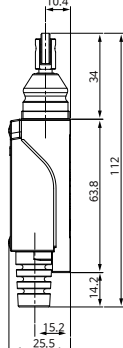
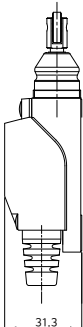
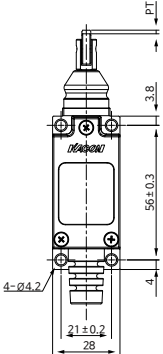
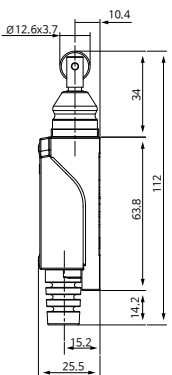
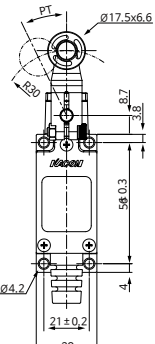
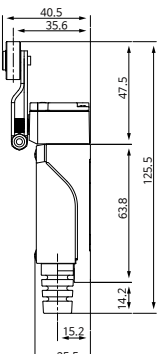
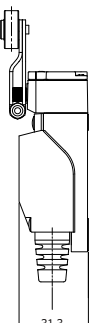
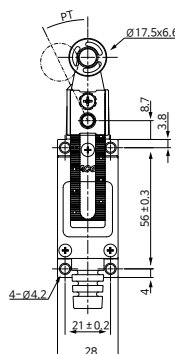
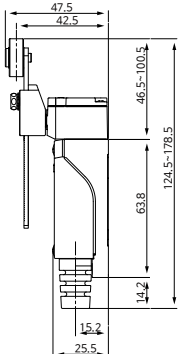
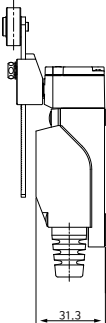
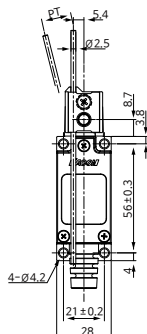
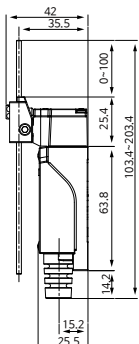
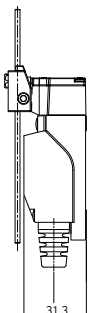
## KXM Series

### Product Selection

|                                                                                     | Indicator Lamp | Part Number | Head Type                              | OF                                     | RF                                   | PT  | MD  | OT  | TT  |
|-------------------------------------------------------------------------------------|----------------|-------------|----------------------------------------|----------------------------------------|--------------------------------------|-----|-----|-----|-----|
|    | Standard       | KXM-726     | Ø50<br>Adjustable Roller<br>Lever Type | 800gf<br>(7.84N)<br>~ 342gf<br>(3.35N) | 50gf<br>(0.49N)<br>~ 21gf<br>(0.21N) | 20° | 10° | 75° | 95° |
|                                                                                     | LED Lamp       | KXM-726D    |                                        |                                        |                                      |     |     |     |     |
|                                                                                     | Neon Lamp      | KXM-726A    |                                        |                                        |                                      |     |     |     |     |
|    | Standard       | KXM-901     | Spring Rod                             | 90gf<br>(0.88N)                        |                                      | 30° |     | 20° | 50° |
|                                                                                     | LED Lamp       | KXM-901D    |                                        |                                        |                                      |     |     |     |     |
|                                                                                     | Neon Lamp      | KXM-901A    |                                        |                                        |                                      |     |     |     |     |
|  | Standard       | KXM-902     | Spring Wire                            | 90gf<br>(0.88N)                        |                                      | 30° |     | 20° | 50° |
|                                                                                     | LED Lamp       | KXM-902D    |                                        |                                        |                                      |     |     |     |     |
|                                                                                     | Neon Lamp      | KXM-902A    |                                        |                                        |                                      |     |     |     |     |
|  | Standard       | KXM-903     | Spring Rod-Plastic Tip                 | 90gf<br>(0.88N)                        |                                      | 30° |     | 20° | 50° |
|                                                                                     | LED Lamp       | KXM-903D    |                                        |                                        |                                      |     |     |     |     |
|                                                                                     | Neon Lamp      | KXM-903A    |                                        |                                        |                                      |     |     |     |     |
|  | Standard       | KXM-923     | Spring Rod-Metal Tip                   | 90gf<br>(0.88N)                        |                                      | 30° |     | 20° | 50° |
|                                                                                     | LED Lamp       | KXM-923D    |                                        |                                        |                                      |     |     |     |     |
|                                                                                     | Neon Lamp      | KXM-923A    |                                        |                                        |                                      |     |     |     |     |

## Dimension

unit : mm

| Push Plunger<br>KXM-301                                                             |                                                                                     |                                                                                     | Roller Plunger<br>KXM-302                                                            |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Standard Type                                                                       | with Neon / LED Type                                                                |                                                                                     | Standard Type                                                                        | with Neon / LED Type                                                                  |                                                                                       |
|    |    |    |    |    |    |
| Cross Roller Plunger<br>KXM-312                                                     |                                                                                     |                                                                                     | Roller Lever<br>KXM-702                                                              |                                                                                       |                                                                                       |
| Standard Type                                                                       | with Neon / LED Type                                                                |                                                                                     | Standard Type                                                                        | with Neon / LED Type                                                                  |                                                                                       |
|   |   |   |   |   |   |
| Adjustable Roller Lever<br>KXM-703                                                  |                                                                                     |                                                                                     | Adjustable Rod Lever<br>KXM-704                                                      |                                                                                       |                                                                                       |
| Standard Type                                                                       | with Neon / LED Type                                                                |                                                                                     | Standard Type                                                                        | with Neon / LED Type                                                                  |                                                                                       |
|  |  |  |  |  |  |

Rev. 2/14

Data subject may change without notice.

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Industrial Controls Catalog

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**KACON**

# Limit Switch

KXM Series

## Dimension

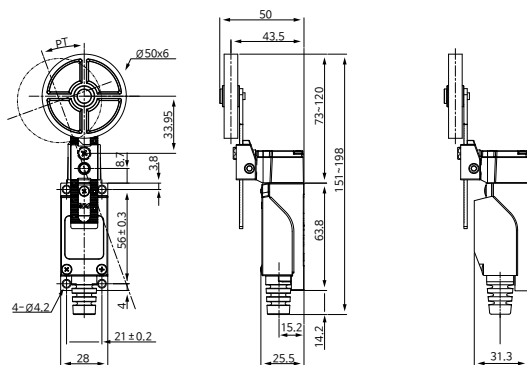
unit : mm

### Ø50 Adjustable Roller Lever Type

KXM-726

Standard Type

with Neon / LED Type

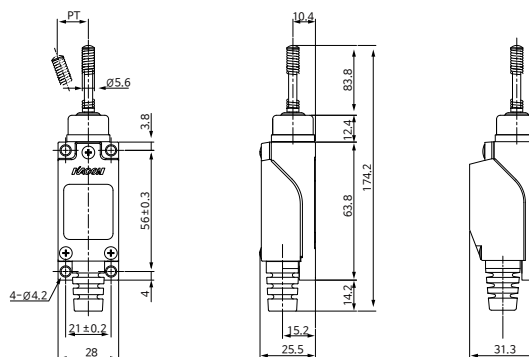


### Spring Rod

KXM-901

Standard Type

with Neon / LED Type

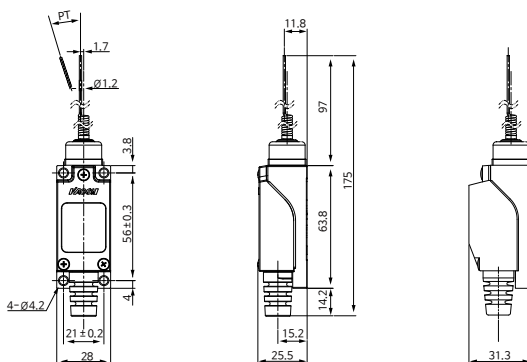


### Spring Wire

KXM-902

Standard Type

with Neon / LED Type

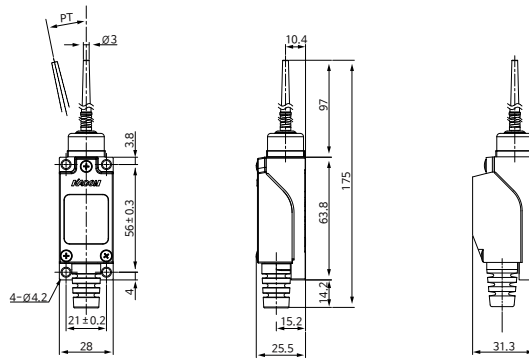


### Spring Rod-Plastic Tip

KXM-903

Standard Type

with Neon / LED Type

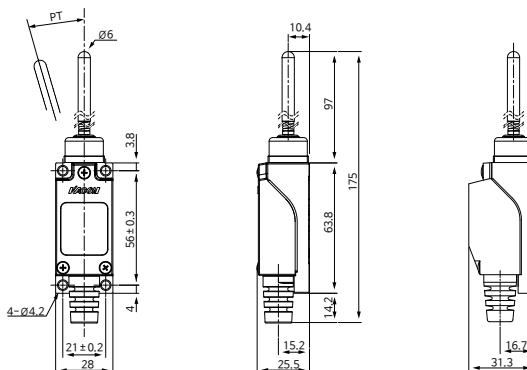


### Spring Rod-Metal Tip

KXM-923

Standard Type

with Neon / LED Type



## Operation indicator lamp Ratings

| Indicator Lamp | Item | Rated voltage | Rated current | Order code        |
|----------------|------|---------------|---------------|-------------------|
| NEON lamp      |      | 80 ~ 240VAC   | 0.2 ~ 1.5mA   | AC load: KXM - □A |
| LED lamp       |      | 12 ~ 24VDC    | 0.5 ~ 1.5mA   | DC load: KXM - □D |

- There are two types of lamps: Neon and LED lamps.
- The circuit check and fault detection are easy because the switch operation status is easily checked outside.
- This is ideal for the places where checking the passage of products is difficult on the conveyor line.
- The LED type has the + or - polarity. Pay attention to the polarity, which is indicated inside.
- The lens section of the indicator lamp cover ensures sufficient brightness with its diffusible resin diamond cut structure.
- The waterproof performance is excellent because the indicator lamp and cover are formed together.



## Operation indicator lamp direction shift

- The lamp terminal does not require additional wiring, because it is connected via the spring in the indicator cover and the built-in switch terminal bolts.
- The indicator lamp can be selected between the modes of "ON in operation" and "ON in nonoperation".
- The indicator lamp can be shifted by removing the lamp holder inside the cover using tools including drivers.



Lamp holder removal



ON in operation

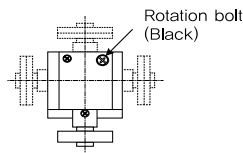


ON in non-operation

- The user can change the mode of operation status indication by changing the lamp holder mounting direction by 180° (up or down) inside the indicator lamp cover.
- The factory setting is "ON in operation"

## Head mounting

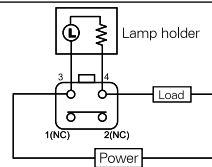
In the lever type, the head can be placed in one of four directions at 90° rotation interval.



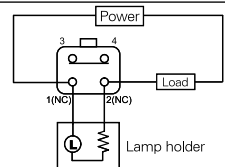
To change the head direction, loosen the black bolt on the top of the head, rotate the head, and tighten the bolt again.

## Operation indicator lamp wiring and circuit diagram

### Internal wiring



ON in non-operation  
[Lamp ON when NC ON]

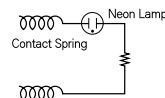


ON in operation  
[Lamp ON when NO ON]

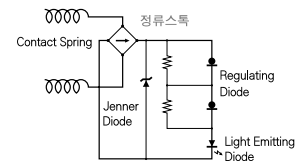
Caution: Use the same power supply for terminals /  
Connect the same polarity for terminals 1 and 4, and 2 and 3.

### Diagram

#### NEON Lamp

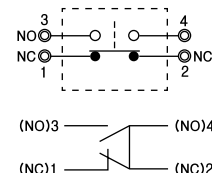
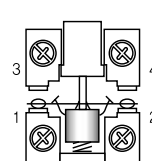


#### LED Lamp



## Built-in switch circuit diagram

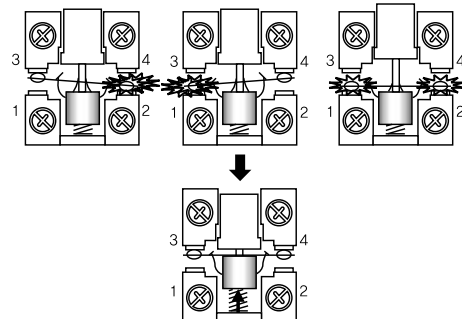
### 24K Au Plate



## Direct contact opening ⇔ Direct Opening

1. The device is safe with the mechanism that cuts off the circuit when the switch contact weld occurs.
2. If the contact weld appears, the plunger directly pushes up the movable terminal to forcefully separate the NC contact and cut off the circuit.
3. The direct opening function is for the Nc contact only.
4. The plunger continues to push the movable terminal during operation, and the circuit is not cut off by mechanical impact.

### Welding



Contact weld After direct opening

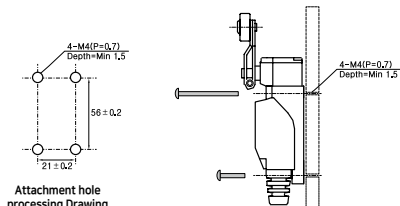
# Limit Switch

## KXM Series

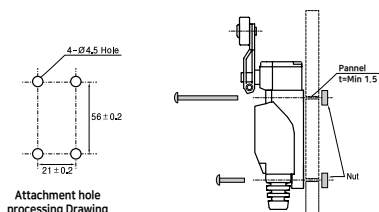
### Mounting on the panel

1. This product can be mounted on the front or rear.
2. Before the installation of the product, determine its position so that the OT of the actuator is proper.  
⇒ An excessive OT leads to a decrease in the product life due to the strong wear and fatigue
3. After determining the mounting position, refer to the following types to process the panel.
4. Be sure to use spring washers for the mounting bolts so that the product will not be removed by vibration or impact.

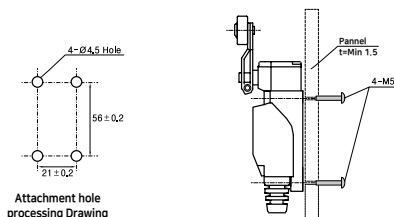
#### Type 1. Front Attached - Direct Tap Processing on the panel



#### Type 2. Front Attachment - Direct hole processing on the panel



#### Type 3. Rear Attachment - Using M5 Tap of the product



☞ The bolts and nuts above are shipped together with the product.

### Wires

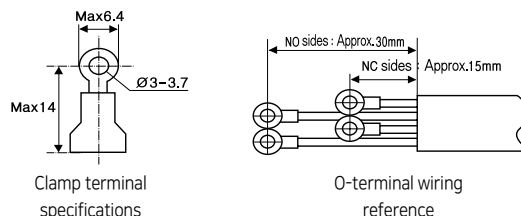
1. The proper size of wires that pass through the wiring lead-out (NBR) is  $\varnothing 6.6 \sim \varnothing 8.5$ .
2. The use of VCTF cable is desirable.
3. The following table shows the specifications of the VCTF cable. Refer to the table for cable selection.

| Small Line Number | Conductor (L)       | Completed outer diameter (I) |
|-------------------|---------------------|------------------------------|
| 2 wire            | 0.75mm <sup>2</sup> | $\varnothing 6.6$ mm         |
|                   | 1.25mm <sup>2</sup> | $\varnothing 7.4$ mm         |
|                   | 2.0mm <sup>2</sup>  | $\varnothing 8.0$ mm         |
| 3 wire            | 0.75mm <sup>2</sup> | $\varnothing 7.0$ mm         |
|                   | 1.25mm <sup>2</sup> | $\varnothing 7.8$ mm         |
|                   | 2.0mm <sup>2</sup>  | $\varnothing 8.5$ mm         |
| 4 wire            | 0.75mm <sup>2</sup> | $\varnothing 7.6$ mm         |
|                   | 1.25mm <sup>2</sup> | $\varnothing 8.5$ mm         |



VCTF : The cable for 300 V or less indoor small appliances, which is manufactured by insulating stranded copper wires with PVC.

4. When wiring the switch terminals, do not directly connect the lead wire to the terminal, but use the O-type clamp terminal with insulation shielding.
5. After wiring, be sure to mount the cover before applying the power. Otherwise, it may lead to an electric shock.



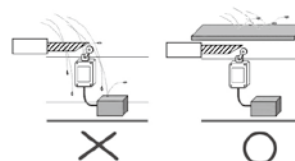
### Tightening torque

When mounting the product, excessive force for bolts or nuts may damage the product. Observe the following tightening torques.

|                  |                              |
|------------------|------------------------------|
| Front Attachment | M4- Max. 12kgf.cm / 1.2N.m   |
| Rear Attachment  | M5- Max. 19.6kgf.cm / 2.0N.m |
| Switch Terminal  | M3- Max. 4.9kgf.cm / 0.5N.m  |

### Environment

- Do not use the product under water or oil, nor in the environment where it is influenced by oil or water because this product is not completely oiltight.
- Do not use the product in the places where there are risks of fire and corrosion.
- Actual conditions may affect the product reliability including the guaranteed life and tightness.

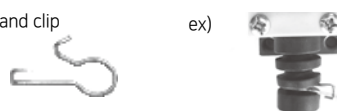


### Basic Accessories

- Front attachment-M4×35 2EA, M4×20 2EA
- Rear Attachment-M5×10 4EA



- Band clip



Tighten the bottom wire outlet using a band clip.